

Deep Dive: 16. OOP I: Classes

Concept Summary

Object-Oriented Programming models real-world entities. A Class is a blueprint; an Object is a physical instance built from it.

Coding Structure

```
class Name:
    def __init__(self):
        self.attr = value
```

Example Code

```
class Dog:
    def __init__(self, name):
        self.name = name
    def bark(self):
        print(f'{self.name} says Woof!')

d1 = Dog('Buddy')
d1.bark()
```

Code Explanation

- Line 1: Defines a class (blueprint) named Dog.
- Line 2: The **init** method is the Constructor. It sets up the object. 'self' refers to the specific physical dog being created.
- Line 4: A function inside a class is called a Method.
- Line 7-8: We instantiate an object (d1) and call its method.

Practice Problems

1. Create a 'Car' class with make and model attributes, and a 'drive' method.
2. Create two different Car objects and make them both drive.